

# ECE 105: Introduction to Electrical Engineering

Lecture 9 & 10

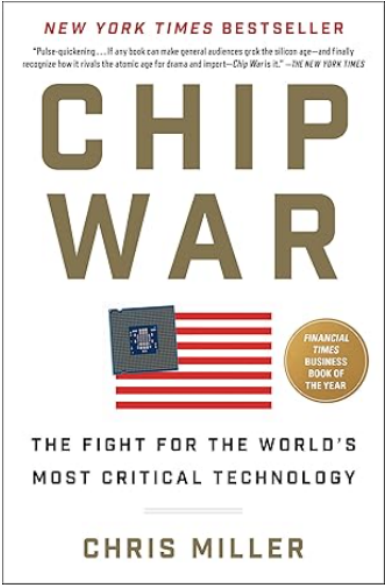
Device 1 & 2

Yasser Khan

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# Chip War



## Chip War: The Fight for the World's Most Critical Technology Paperback

– September 16, 2025

by [Chris Miller](#) (Author)

4.7 ★★★★★ (8,924) 4.4 on Goodreads 37,925 ratings

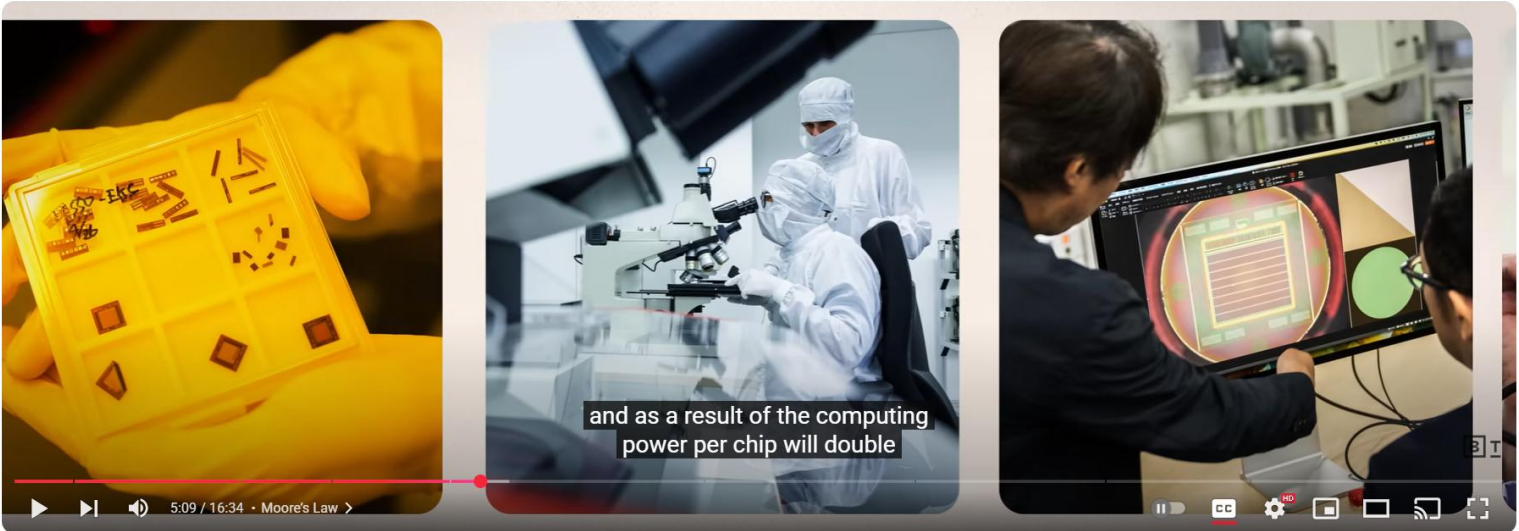
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**The *Financial Times* Business Book of the Year, this epic account of the decades-long battle to control one of the world's most critical resources—microchip technology—with the United States and China increasingly in fierce competition is “pulse quickening...a nonfiction thriller” (*The New York Times*).**

You may be surprised to learn that microchips are the new oil—the scarce resource on which the modern world depends. Today, military, economic, and geopolitical power are built on a foundation of computer chips. Virtually *everything*—from missiles to microwaves—runs on chips, including cars, smartphones, the stock market, even the electric grid. Until recently, America designed and built the fastest chips and maintained its lead as the #1 superpower, but America's edge is in danger of slipping, undermined by players in Taiwan, Korea, and China, which spends more on chips than any other power in the world. At stake is America's military superiority and economic dominance.

Economic historian Chris Miller explains how the semiconductor industry has shaped the modern world. [Read more](#)

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How an AI chip war could destroy the global economy | Chris Miller for The Freethink Interview



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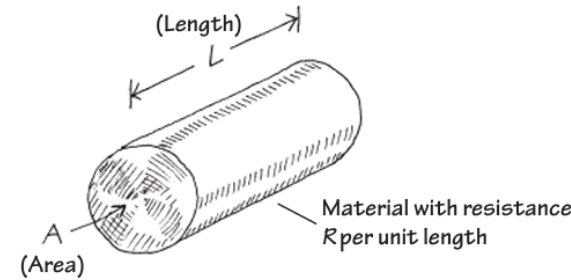
<https://www.youtube.com/watch?v=foYWzdvajvo>

# How can we create tunable resistors?

- $R = \rho L/A$
- We have three options, change length, change area, or change resistivity
- Semiconductors allow us to electromagnetically and chemically control resistivity

# How do semiconductors allow us to tune the resistivity?

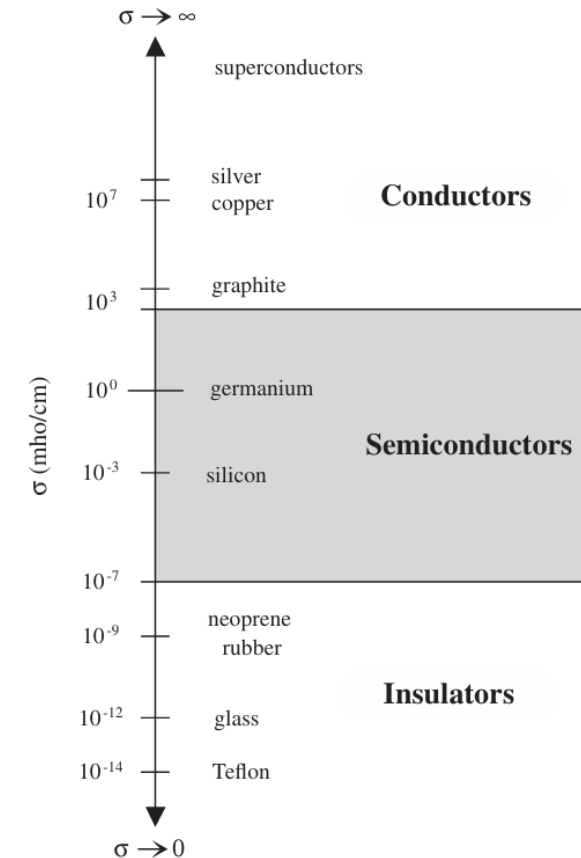
- By allowing us to tune the density of charge carriers!
- In metals, the density of charge carriers is  $\sim 10^{22} - 10^{23}/\text{cm}^3$
- In insulators, the charge density is  $\sim 10^{-57}/\text{cm}^3$
- In semiconductors, the charge density can be controlled from  $\sim 10^{10}/\text{cm}^3 - 10^{20}/\text{cm}^3$



$$\rho = R \frac{A}{L} \text{ (Resistivity ohm}\cdot\text{cm)}$$

$$\sigma = \frac{1}{\rho} \text{ (Conductivity mho/cm)}$$

$$\text{mho} = \frac{1}{\text{ohm}} = \frac{1}{\Omega} = \mathcal{O}$$



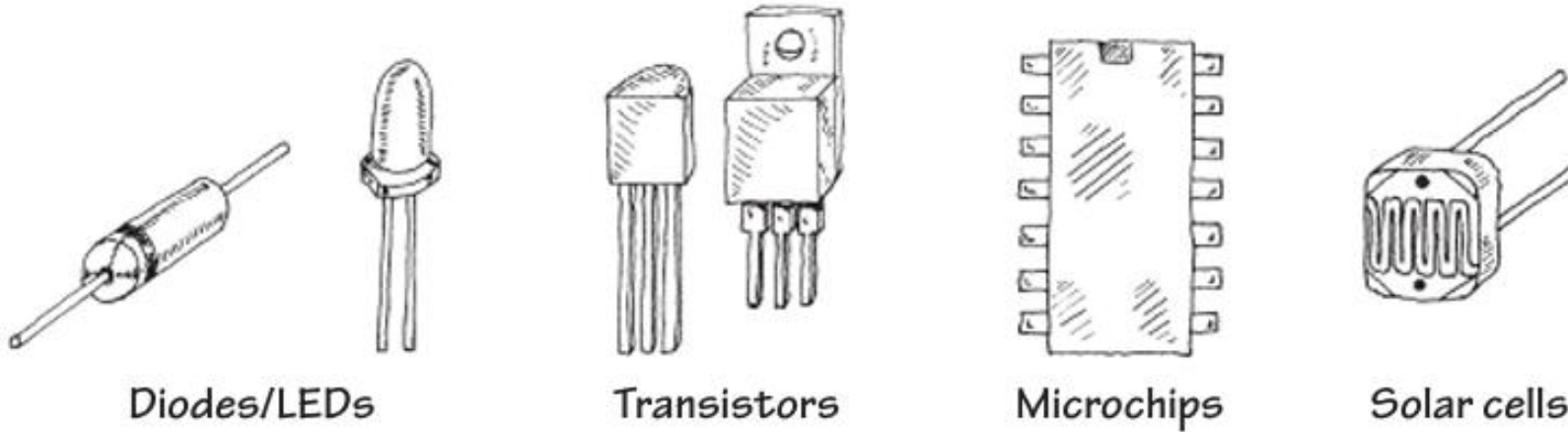
# What can we do with an element that has tunable or non-linear resistance?

- Create a switch
  - Digital logic
- Amplify a signal
  - Analog circuits
- Rectify an AC voltage
  - Power circuits
- Sensing, power generation, light, etc

# Why does changing the carrier density change the resistivity?

- Let's revisit  $R = \rho L/A$
- If I take a material and magically stuff additional mobile electrons into it, then the resistivity would decrease.
- If I remove additional mobile electrons from it, then the resistivity would increase.
- So if I had a material which allows me to change the mobile charge density, then I can tune the resistivity of that material

# What's after resistors, capacitors, and inductors



- Each device and circuit allows us to carry out specific functionalities

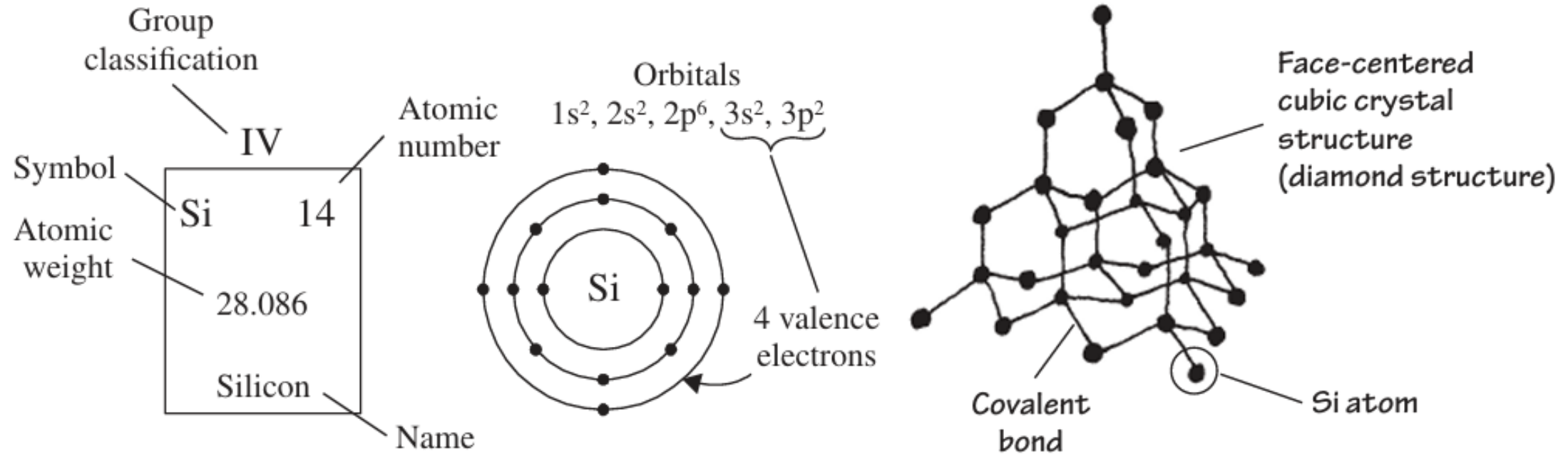


**USC**  
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# Periodic Table of the Elements

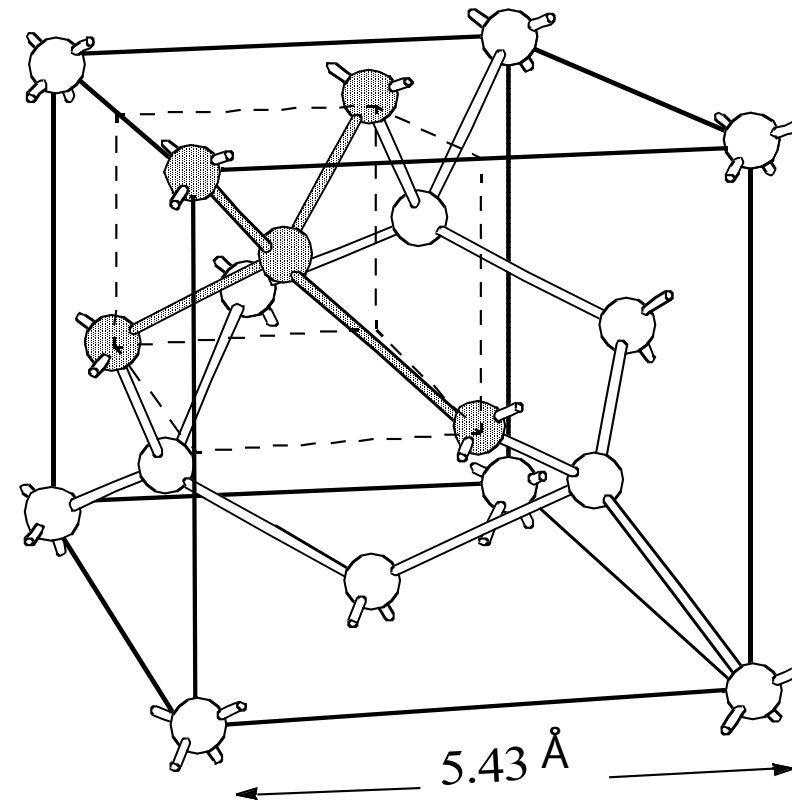
Binary: GaAs, SiC, CdSe, etc.  
Ternary+: AlGaAs, InGaAs, etc.

# Silicon – the workhorse of electronics

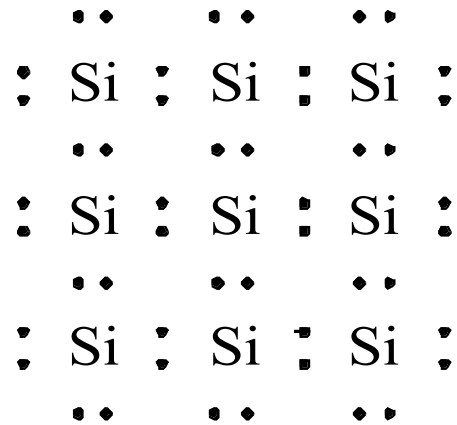


## Silicon Crystal Structure

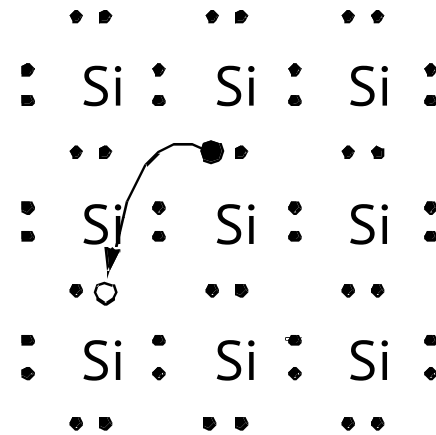
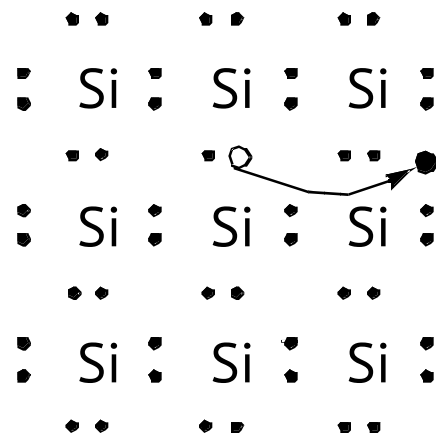
- ***Unit cell*** of silicon crystal is cubic.
- ***Each Si atom has 4 nearest neighbors.***



# Bond Model of Electrons and Holes (Intrinsic Si)

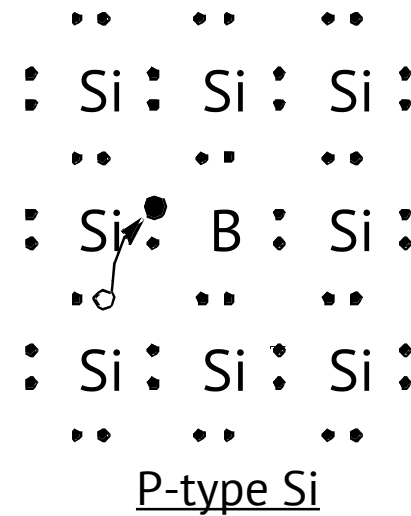
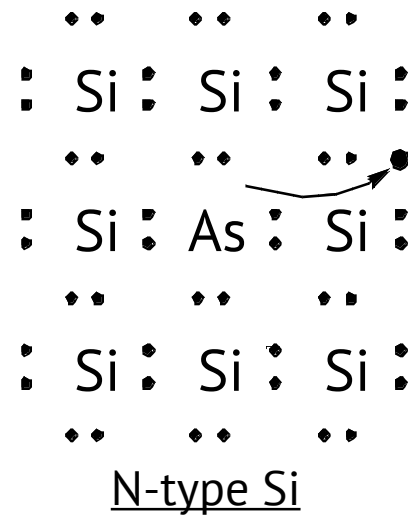


- Silicon crystal in a two-dimensional representation.



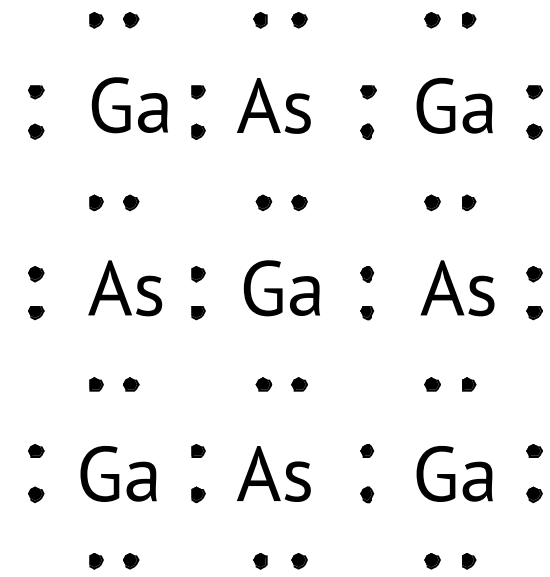
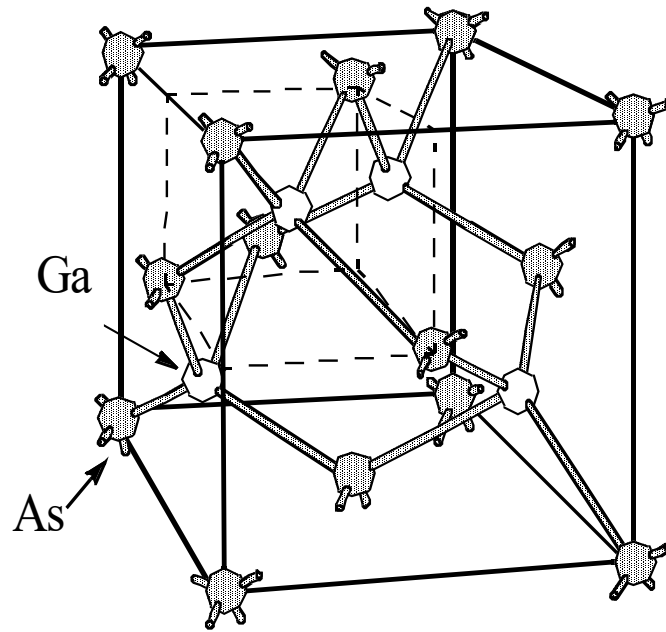
- When an electron breaks loose and becomes a **conduction electron**, a **hole** is also created.

# Dopants in Silicon



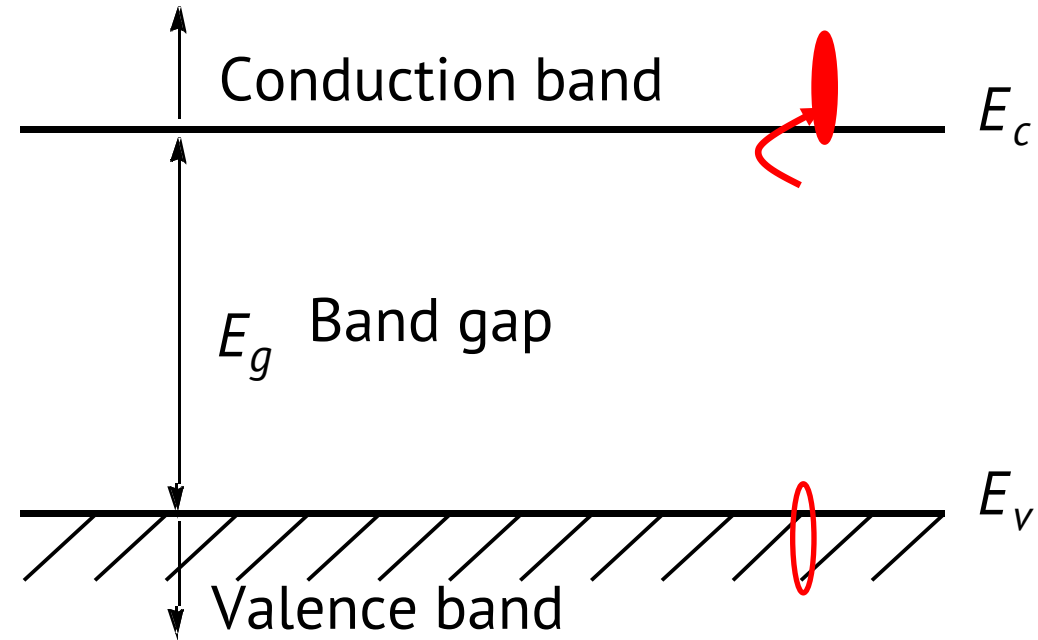
- As (Arsenic), a Group V element, introduces conduction electrons and creates ***N-type silicon***, and is called a ***donor***.
- B (Boron), a Group III element, introduces holes and creates ***P-type silicon***, and is called an ***acceptor***.
- Donors and acceptors are known as dopants.

# GaAs, III-V Compound Semiconductors, and Their Dopants



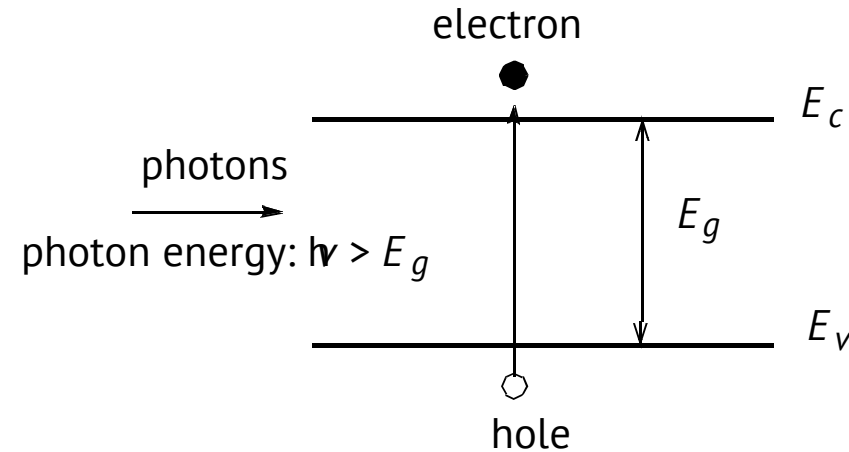
- GaAs has the same crystal structure as Si.
- GaAs, GaP, GaN are III-V compound semiconductors, important for optoelectronics.
- Which group of elements are candidates for donors? acceptors?

# Energy Band Diagram



- **Energy band diagram** shows the bottom edge of conduction band,  $E_c$ , and top edge of valence band,  $E_v$ .
- $E_c$  and  $E_v$  are separated by the **band gap energy,  $E_g$** .

# Measuring the Band Gap Energy by Light Absorption



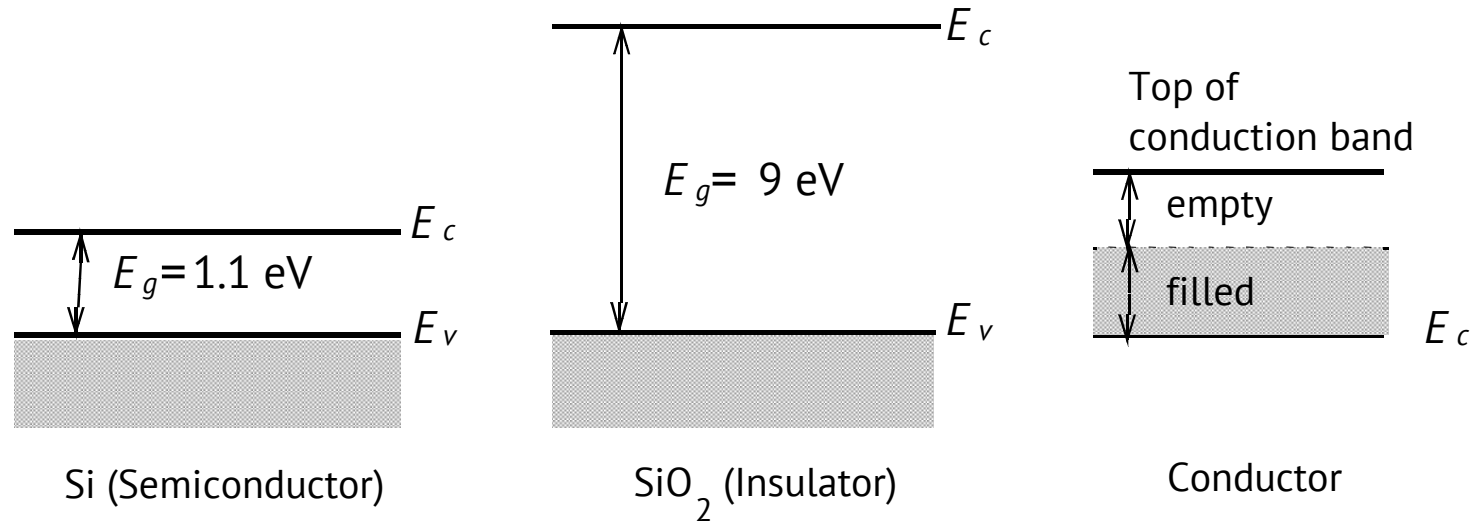
- $E_g$  can be determined from the minimum energy ( $h\nu$ ) of photons that are absorbed by the semiconductor.

**Bandgap energies of selected semiconductors**

| Material   | PbTe | Ge   | Si   | GaAs | GaP  | Diamond |
|------------|------|------|------|------|------|---------|
| $E_g$ (eV) | 0.31 | 0.67 | 1.12 | 1.42 | 2.25 | 6.0     |



# Semiconductors, Insulators, and Conductors



- Totally filled bands and totally empty bands do not allow current flow. (Just as there is no motion of liquid in a totally filled or totally empty bottle.)
- Metal conduction band is half-filled.
- Semiconductors have lower  $E_g$ s than insulators and can be doped.

# Resistance calculation and changing by doping

## Example A — Intrinsic silicon (no intentional doping)

- $n = p = n_i = 10^{10} \text{ cm}^{-3}$
- $\sigma = q(\mu_n + \mu_p)n_i \approx 1.602 \times 10^{-19} \times (1350 + 480) \times 10^{10} \text{ S/cm}$
- $\rho \approx 3.41 \times 10^5 \Omega \cdot \text{cm}$

If you have a bar  $L = 1 \text{ cm}$ ,  $A = 1 \text{ mm}^2 = 0.01 \text{ cm}^2$ :

$$R = \rho L / A \approx 3.41 \times 10^5 \times (1/0.01) = 3.41 \times 10^7 \Omega \text{ } (\approx 34 \text{ M}\Omega)$$

# Resistance calculation and changing by doping

**Example B — Light n-type doping  $N_D = 10^{15} \text{ cm}^{-3}$**

- Majority electrons  $n \approx N_D = 10^{15}$ , minority holes  $p \approx n_i^2/n \approx 10^{20}/10^{15} = 10^5$  (negligible)
- Use low-doping mobilities:  $\mu_n \approx 1350$ ,  $\mu_p \approx 480$
- $\sigma \approx q \mu_n n \Rightarrow \rho \approx 4.62 \Omega \cdot \text{cm}$

Bar ( $1 \text{ cm} \times 1 \text{ mm}^2$ ):  $R \approx 4.62 \times (1/0.01) = \mathbf{462 \Omega}$

$1 \mu\text{m}$  film:  $R_s \approx \mathbf{46.2 \text{ k}\Omega/\square}$

**Example C — Moderate n-type  $N_D = 10^{17} \text{ cm}^{-3}$  with mobility degradation**

- $n \approx 10^{17}$ ,  $p \approx 10^{20}/10^{17} = 10^3$  (negligible)
- Mobility drops with doping; take  $\mu_n \approx 1000 \text{ cm}^2/\text{V s}$  (typical),  $\mu_p \approx 480$
- $\sigma \approx q \mu_n n \Rightarrow \rho \approx 6.24 \times 10^{-2} \Omega \cdot \text{cm}$

Bar ( $1 \text{ cm} \times 1 \text{ mm}^2$ ):  $R \approx \mathbf{6.24 \Omega}$

$1 \mu\text{m}$  film:  $R_s \approx \mathbf{624 \Omega/\square}$

# Resistance calculation and changing by doping

**Example C — Moderate n-type  $N_D = 10^{17} \text{ cm}^{-3}$  with mobility degradation**

- $n \approx 10^{17}$ ,  $p \approx 10^{20}/10^{17} = 10^3$  (negligible)
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Bar ( $1 \text{ cm} \times 1 \text{ mm}^2$ ):  $R \approx \mathbf{6.24 \Omega}$

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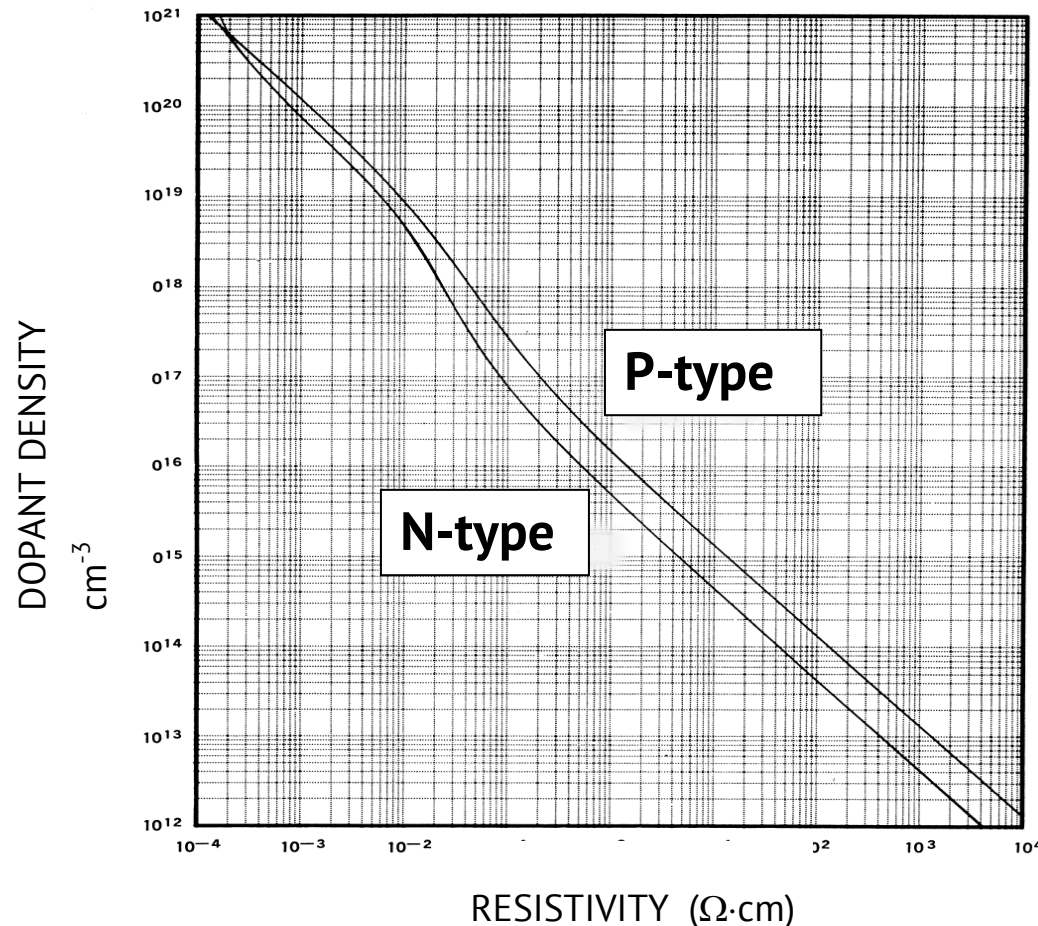
**Example D — Heavy p-type  $N_A = 10^{18} \text{ cm}^{-3}$  with mobility degradation**

- $p \approx 10^{18}$ ,  $n \approx 10^{20}/10^{18} = 10^2$  (negligible)
- Hole mobility degrades more; take  $\mu_p \approx 200 \text{ cm}^2/\text{V s}$ ,  $\mu_n \approx 1350$  (minority)
- $\sigma \approx q \mu_p p \Rightarrow \rho \approx 3.12 \times 10^{-2} \Omega \cdot \text{cm}$

Bar ( $1 \text{ cm} \times 1 \text{ mm}^2$ ):  $R \approx \mathbf{3.12 \Omega}$

$1 \mu\text{m}$  film:  $R_s \approx \mathbf{312 \Omega/\square}$

# Resistance calculation and changing by doping



$$\rho = 1/\sigma$$

(a) What is the resistivity ( $\rho$ ) of silicon doped with  $10^{17}\text{cm}^{-3}$  of arsenic?

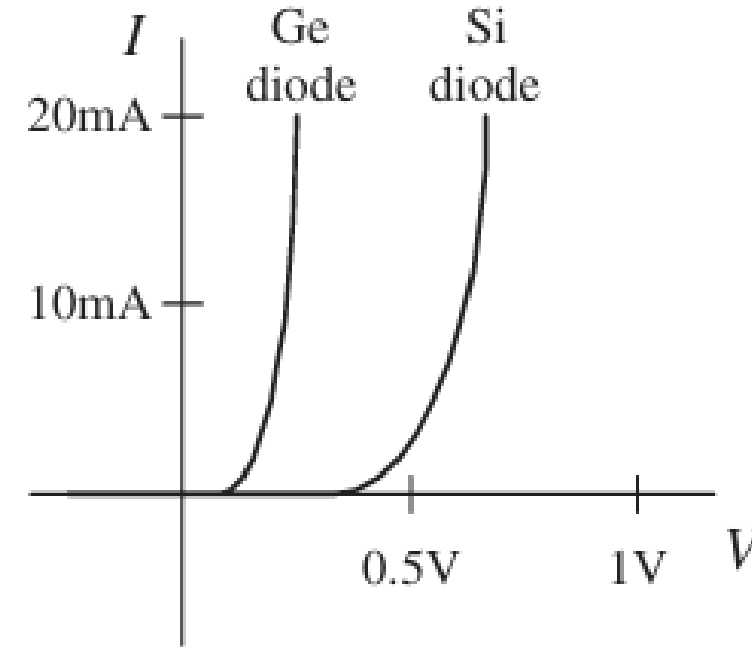
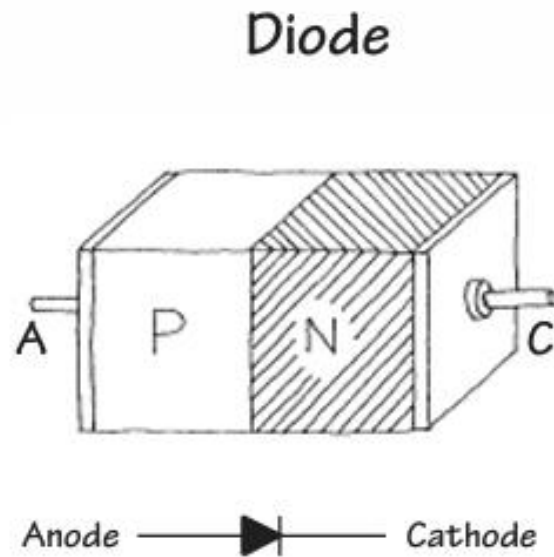
(b) What is the resistance ( $R$ ) of a piece of this silicon material  $1\mu\text{m}$  long and  $0.1\mu\text{m}^2$  in cross-sectional area?

## Solution:

(a) Using the N-type curve in the previous figure, we find that  $\rho = 0.084\Omega\cdot\text{cm}$ .

$$\begin{aligned} \text{(b) } R &= \rho L/A = 0.084\Omega\cdot\text{cm} \times 1\mu\text{m} / 0.1\mu\text{m}^2 \\ &= 0.084\Omega\cdot\text{cm} \times 10^{-4}\text{cm} / 10^{-10}\text{cm}^2 \\ &= 8.4 \times 10^{-4}\Omega \end{aligned}$$

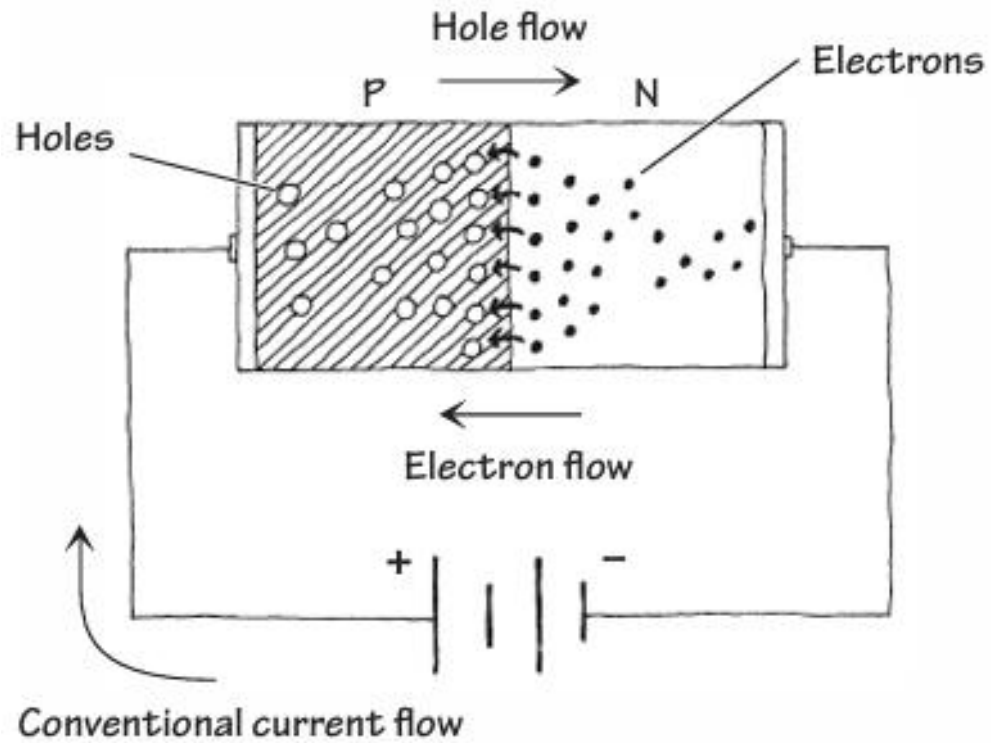
# What is a diode



- A diode is a device that exhibits 'rectifying' behavior
- Can be made from a physical junction between an n-type semiconductor and a p-type semiconductor
- Can also be made from a physical junction between a metal and semiconductor (not discussed here)

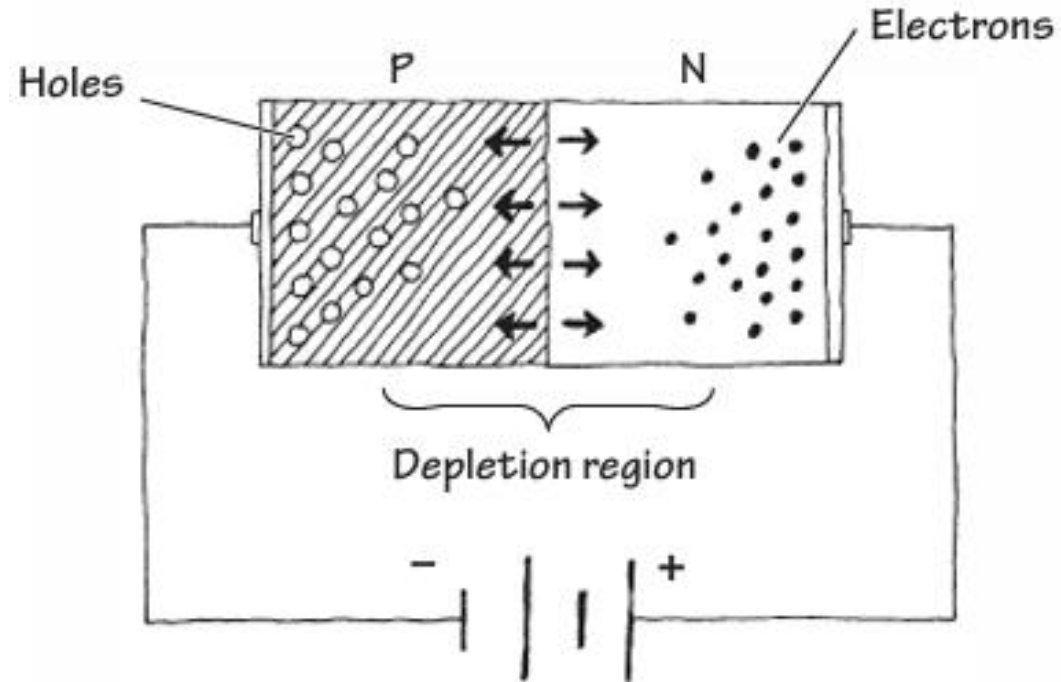
# Forward biasing a diode

## Forward-Biased ("Open Door")



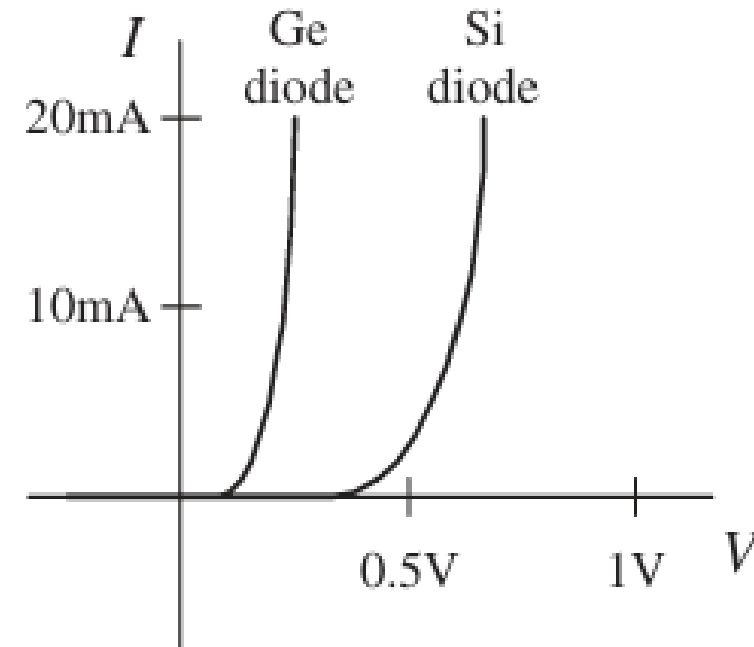
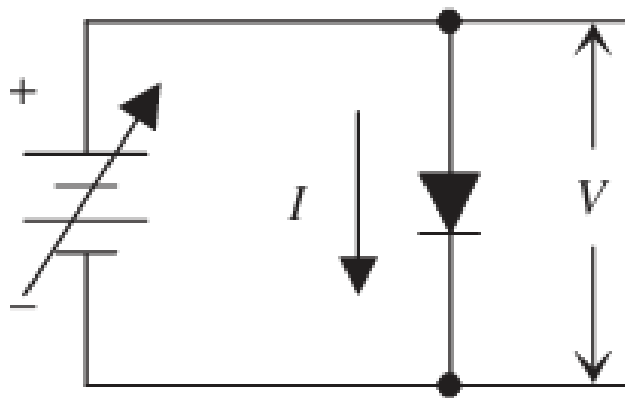
# Reverse biasing a diode

## Reverse-Biased ("Closed Door")

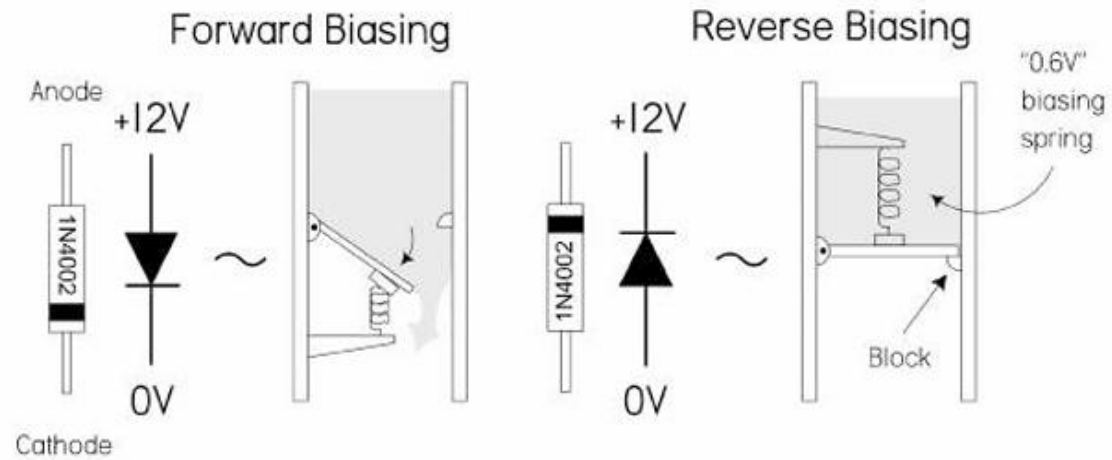




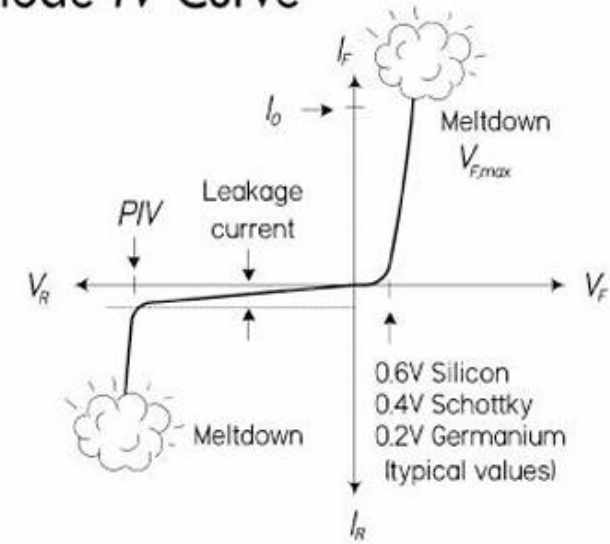
# Diode characterization



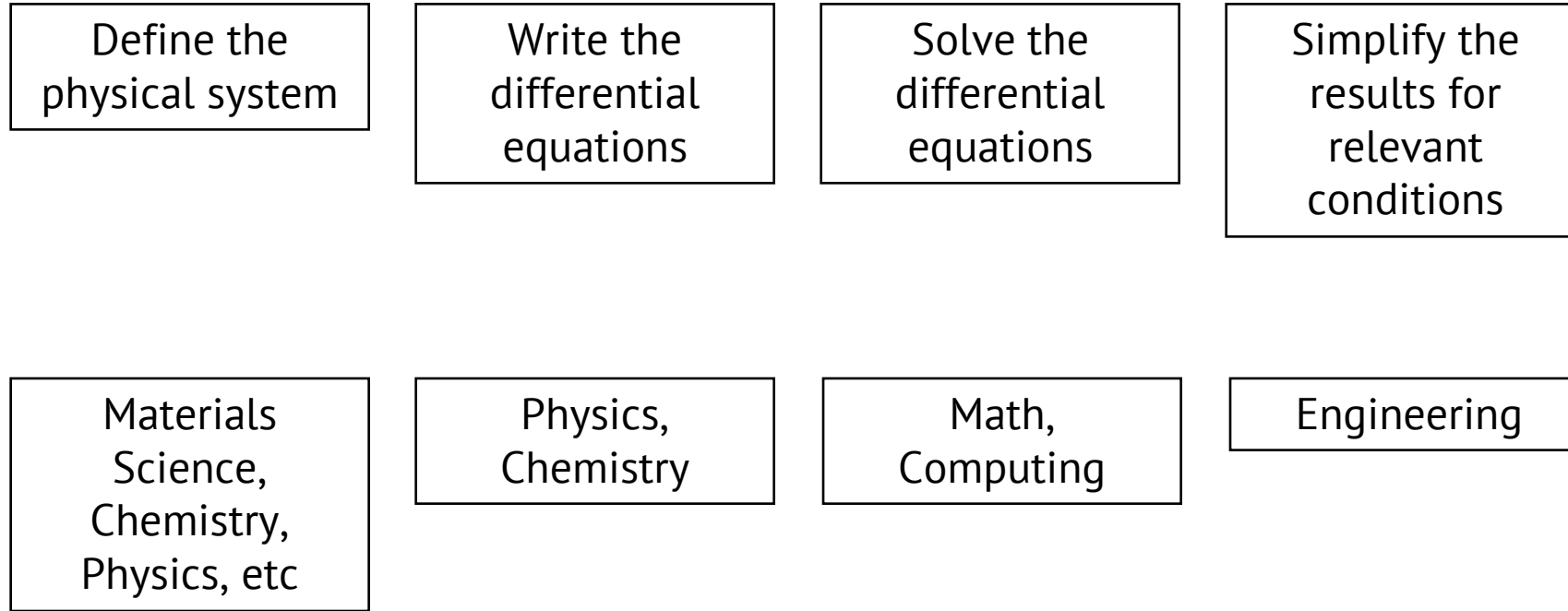
## Diode Water Analogy



## Diode IV Curve



# How do we figure out the behavior?

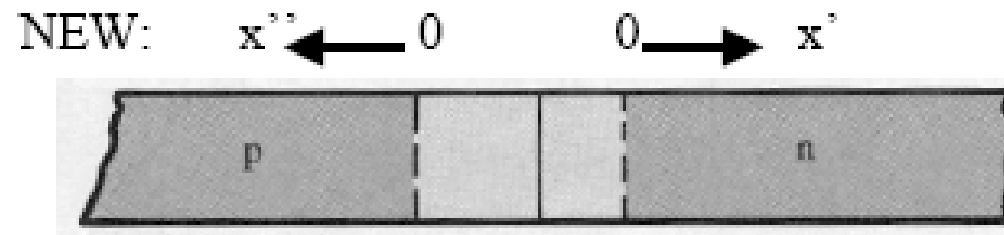


# Solving for diode current equations

- From the minority carrier diffusion equation:  $\frac{d^2 \Delta p_n}{dx^2} = \frac{\Delta p_n}{D_p \tau_p} = \frac{\Delta p_n}{L_p^2}$
- We have the following boundary conditions:

$$\Delta p_n(x_n) = p_{no} (e^{qV_A/kT} - 1) \quad \Delta p_n(\infty) \rightarrow 0$$

- For simplicity, we will develop a new coordinate system:



- Then, the solution is of the form:

$$\Delta p_n(x') = A_1 e^{x'/L_p} + A_2 e^{-x'/L_p}$$

# Solving for diode current equations

$$\Delta p_n(x') = A_1 e^{x'/L_p} + A_2 e^{-x'/L_p}$$

From the  $x = \infty$  boundary condition,  $A_1 = 0$ .

From the  $x = x_n$  boundary condition,  $A_2 = p_{no}(e^{qV_A/kT} - 1)$

Therefore,  $\Delta p_n(x') = p_{no}(e^{qV_A/kT} - 1)e^{-x'/L_p}$ ,  $x' > 0$

Similarly, we can derive

$$\Delta n_p(x'') = n_{po}(e^{qV_A/kT} - 1)e^{-x''/L_n}, \quad x'' > 0$$

# Solving for diode current equations

- Current density  $J = J_n(x) + J_p(x)$

$$J_n(x) = q\mu_n n \mathcal{E} + qD_n \frac{dn}{dx} = q\mu_n n \mathcal{E} + qD_n \frac{d(\Delta n)}{dx}$$

$$J_p(x) = q\mu_p p \mathcal{E} - qD_p \frac{dp}{dx} = q\mu_p p \mathcal{E} - qD_p \frac{d(\Delta p)}{dx}$$

- $J$  is constant throughout the diode, but  $J_n(x)$  and  $J_p(x)$  vary with position

# Solving for diode current equations

$$\textbf{p-side: } J_n = -qD_n \frac{d\Delta n_p(x'')}{dx''} = q \frac{D_n}{L_n} n_{p0} (e^{qV_A/kT} - 1) e^{-x''/L_n}$$

$$\textbf{n-side: } J_p = -qD_p \frac{d\Delta p_n(x')}{dx'} = q \frac{D_p}{L_p} p_{n0} (e^{qV_A/kT} - 1) e^{-x'/L_p}$$

$$J = J_n \Big|_{x=-x_p} + J_p \Big|_{x=x_n} = J_n \Big|_{x''=0} + J_p \Big|_{x'=0}$$

$$J = qn_i^2 \left[ \frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D} \right] (e^{qV_A/kT} - 1)$$

# Ideal Diode Current Equation

$$J = qn_i^2 \left[ \frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D} \right] (e^{qV_A/kT} - 1) \longrightarrow I = I_0 (e^{\frac{qV}{kT}} - 1)$$

- This equation describes the behavior of an ideal diode



$$I = I_0 \left( e^{\frac{qV}{n k T}} - 1 \right)$$

- This equation partially describes the behavior of a non-ideal diode

# Non-Ideal Diode Current Equation

$$I = I_s \left( e^{\frac{V}{nV_T}} - 1 \right)$$

where:

- $I$  = diode current
- $I_s$  = reverse saturation current (very small, e.g.  $10^{-12}$  A)
- $V$  = applied voltage across the diode
- $n$  = ideality factor ( $\approx 1$  for silicon)
- $V_T = \frac{kT}{q} \approx 25.9$  mV at 300K

---

## Example 1 — Forward bias at 0.7 V

- Let  $I_s = 10^{-12}$  A,  $n = 1$ ,  $V_T = 0.026$  V
- $V = 0.7$  V

$$I = 10^{-12} \left( e^{\frac{0.7}{0.026}} - 1 \right)$$

$$\frac{0.7}{0.026} \approx 26.9 \quad \Rightarrow \quad e^{26.9} \approx 5.3 \times 10^{11}$$

$$I \approx 10^{-12} \times 5.3 \times 10^{11} \approx 0.53 \text{ A}$$

# Non-Ideal Diode Current Equation

Example 2 — Forward bias at 0.6 V

$$I = 10^{-12} \left( e^{\frac{0.6}{0.026}} - 1 \right)$$

$$\frac{0.6}{0.026} \approx 23.1, \quad e^{23.1} \approx 1.1 \times 10^{10}$$

$$I \approx 10^{-12} \times 1.1 \times 10^{10} = 0.011 \text{ A} = 11 \text{ mA}$$

Example 3 — Reverse bias at -5 V

$$I = 10^{-12} \left( e^{\frac{-5}{0.026}} - 1 \right)$$

$$\frac{-5}{0.026} \approx -192 \quad \Rightarrow \quad e^{-192} \approx 0$$

$$I \approx -10^{-12} \text{ A}$$

Example 4 — Small voltage 0.1 V

$$I = 10^{-12} \left( e^{\frac{0.1}{0.026}} - 1 \right)$$

$$\frac{0.1}{0.026} \approx 3.85, \quad e^{3.85} \approx 47$$

$$I \approx 10^{-12} \times (47 - 1) \approx 4.6 \times 10^{-11} \text{ A}$$